DAY 04 / 11 · WEB DEV IN 11 DAYS · GUIDED

JavaScript: first steps

Values, variables, maths: your first instructions for the browser

ShowShelf after today: **JavaScript fills in the title, the facts and how long it takes to watch every episode**

TODAY

The plan for the next 90 minutes

Minutes	Topic
10	Recap of Day 3
13	The Console, text and numbers
10	Variables: const and let
13	Maths and sentences
8	Put text on the page
11	True, false, if and else
5	Read the error
5	The big idea, and questions
15	This afternoon in the Lab

RECAP · DAY 3

Five quick questions

1. Logo and menu in a row: which box gets display: flex?
2. justify-content or align-items: which works along the row?
3. What does flex: 1 on the search input do?
4. What does repeat(3, 1fr) make?
5. Mobile first: who are the normal rules for?

JAVASCRIPT · STEP 1

Instructions, top to bottom

HTML: what things are. CSS: how they look. **JavaScript**: what to do.

The **Console**, under the preview, is where JavaScript talks back to you. In a normal browser: F12.

Capital letters matter: console, not Console.

```
console.log("Good morning!");
console.log("Then the next line.");

// Good morning!
// Then the next line.
```

JAVASCRIPT · STEP 2

Text and numbers

A **string** is text, in quotes. A **number** has no quotes. Decimals use a dot: 8.6.

+ adds numbers, but joins strings.

```
console.log(26 + 10);
// 36

console.log("26" + 10);
// 2610

console.log("The" + " Wire");
// The Wire
```

JAVASCRIPT · STEPS 3-4

A name for a value

= means put into. Use the name **without** quotes.

const: never changes. **let**: can change. Write let only once.

Names in camelCase: **showName**, **seasonsWatched**.

```
const showName = "The Office";
const episodes = 201;
const runtime = 30;

let seasonsWatched = 2;
seasonsWatched = seasonsWatched + 1;
console.log(seasonsWatched);
// 3

// with const instead of let:
TypeError: Assignment to
constant variable.
```

JAVASCRIPT · STEP 5

Maths with variables

+ - * /: plus, minus, times, divided by. Brackets first.

Math.round( ): the nearest whole number. Capital M.

Change one variable and every result that uses it updates.

```
const seasons = 9;
const perSeason =
  episodes / seasons;

console.log(perSeason);
// 22.333333333333332

console.log(Math.round(perSeason));
// 22
```

Sentences with backticks

```
const line = `${showName} has ${seasons} seasons.`;
console.log(line);
// The Office has 9 seasons.
```

Backticks

Around the whole text. Key: top left, under Esc.

${}

A value goes in here: the name of a variable.

Quotes?

Normal quotes print ${showName} as it is.

JAVASCRIPT · STEP 7

Put text on the page

```
document.querySelector("#rating").textContent = "Rating: 8.6";
```

document

The web page

querySelector

Finds the element. The # works like in CSS.

textContent

Sets its text. Capital C.

Today's ids: **title**, **meta**, **watch-time**, **verdict**, **network**, **rating**, **random**

JAVASCRIPT · STEPS 8–9

True, false, if and else

Comparisons give **true** or **false**: > < >= <= === !==

=== checks the type too: 9 and "9" are different.

Exactly one of the two blocks runs.

```
const isShort = runtime < 45;
console.log(isShort);
// true

if (isShort) {
  console.log("Fits in a lunch break.");
} else {
  console.log("One for the evening.");
}
```

JAVASCRIPT · STEP 10

Read the error. It says what and where.

```
Uncaught ReferenceError: episods is not defined
```

Kind of error	For example	What runs
Can't read the code	A missing quote or }	Nothing in the file
Fails while running	A misspelt name	The lines above it

The line number is where JavaScript noticed it. Also look at the line above.

EQUALS SIGNS

One = puts a value into a box. Three === ask if two values are the same.

THIS AFTERNOON IN THE LAB

Ten steps: the page fills itself in

First lines

1 Hello, Console

2 Text and numbers

Variables, maths

3 Variables with const

4 Values that change: let

5 Maths with variables

6 Sentences with backticks

The page

7 Put text on the page

Decisions, errors

8 True or false

9 Choose with if and else

10 Read the error, fix the code

Milestone: your JavaScript fills in the title, the facts, the watch time and an if / else message

Finished early? Extra challenges B1 to B3: text that knows tricks, a random episode, three choices with else if.

STUCK? IN THIS ORDER

Message, Console, hint, answer, ask

1

Message

Read the “Not yet” message. It says what to fix.

2

Console

The red line under the preview: what went wrong, and the line number.

3

Hint

Two per step. The second is nearly the answer.

4

Answer

Compare it with yours. Find the difference.

5

Ask

The teacher, or the person next to you.

END OF THE DAY

Can you answer these?

1. What does "236" + 62 give, and why?
2. Episodes you've watched: const or let?
3. = or ===: which one asks a question?

TOMORROW

Functions and decisions

Package your code into functions you can run again and again, with different numbers each time.